Brickwell Pvt. Ltd.

Construction Project Management System

Comprehensive Requirements Audit, Change Impact Analysis & Company Approval Report

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Document Type: Formal Change Impact Audit Report	**Audit Status:** STRICTLY READ-ONLY (No Code/Data Altered)
Prepared For: Company Owner & Management (Brickwell Pvt. Ltd.)	**Date:** October 2, 2026
Author: Senior Lead Systems Architect & AI Pair Programmer	**Target Stack:** React 19, Vite, TanStack Router, Supabase RLS

CRITICAL AUDIT DIRECTIVE & ASSURANCE:
This document constitutes a thorough, non-destructive audit of the requested application updates. In full compliance with Phase 6 execution rules, ***NO SOURCE CODE, DATABASE SCHEMAS, RLS POLICIES, FINANCIAL CALCULATIONS, OR PRODUCTION RECORDS HAVE BEEN ALTERED****. Implementation remains frozen pending formal management approval.*

1. Executive Summary

Following feedback from company management at Brickwell Pvt. Ltd., this technical audit was commissioned to evaluate six major system update requests and a comprehensive overhaul of the labour attendance and wage tracking module. The primary goal is to ensure that requested enhancements—such as interactive dashboard drill-downs, role-based financial privacy, payroll restructuring, and detailed expense breakdowns—are implemented without compromising database integrity, financial accuracy, auditability, or existing business logic.

Key Takeaways of the Audit:

- **High Architectural Integrity:** The current application leverages a modern React 19 + TanStack Router frontend paired with a robust Supabase PostgreSQL backend enforced by Row-Level Security (RLS). Changes must align with this decoupled architecture.
- **Financial Privacy Gap Identified:** Sensitivity analysis reveals that dashboard financial figures (Contract Value, 6-Month Spend, Cash on Hand) are currently visible to all authenticated roles (`viewer`, `supervisor`, `accounts`, `md`). Restricted role-based access requires backend RLS updates in addition to UI conditional rendering.
- **Payroll Removal Risk:** Deleting the Payroll module entirely would eliminate monthly compliance records, statutory deduction tracking (PF/ESIC), payslip generation, and formal MD approval workflows. We strongly recommend an **integrated Attendance-Payroll model** over complete deletion.
- **Labour Attendance Standardized:** The proposed weekly Attendance and Wage Register perfectly matches the Brickwell reference sheet (13.09.2026 to 19.09.2026), supporting daily worker time tracking, fractional units, team summaries, allowances, advances, and dynamic balances.

2. Existing System Overview

A comprehensive code and schema inspection of the codebase (Corpus: Ajaykannagit/track-site) established the actual application architecture across all major functional areas:

Module / Domain	Source Files & Code Locations	Database Tables & Helper Functions	Current Operational Logic & Roles
Auth & Roles	src/lib/auth.tsx src/routes/_authenticated/route.tsx	user_roles table public.can_write_finance() public.can_write_site()	Roles: md, supervisor, accounts, viewer. Gate checks session; RLS policies govern SQL mutations.
Dashboard & P&L;	src/routes/_authenticated/dashboard .tsx src/routes/_authenticated/reports.tsx	projects, expenses, attendance, contractor_bills, cash_closing, v_project_summary	Computes live project values, 6-month site spend, cash on hand, monthly cost trends, and project-wise profit margins.
Labour & Attendance	src/routes/_authenticated/attendance .tsx src/routes/_authenticated/payroll.tsx	attendance, employees, payroll, payroll_items	Logs daily working hours, OT, status (present/absent/half_day/leave). Payroll stores monthly gross, deductions, net pay.
Procurement & Supply	src/routes/_authenticated/materials.t sx	suppliers, material_requests, purchase_orders, material_receipts, supplier_payments	Full procurement pipeline: Indent (MR) -> Purchase Order (PO) -> Material Receipt (GRN) -> Supplier Payment.
Subcontractors	src/routes/_authenticated/contractors .tsx	contractors, work_orders, measurement_books, contractor_bills, contractor_payments	Tracks piece-rate/lump-sum work orders (WO), physical measurements (MB), net billing (CB), and disbursements.
Accounts & Cash	src/routes/_authenticated/accounts.t sx	expenses, expense_categories, cash_closing, account_transfers	Records site consumables & overheads, daily cash closings (petty cash vs bank account), and account transfers.

3. Individual Requirement Assessments

UPDATE 1 — Dashboard Active Projects and Contract Value

Current State: Active Projects count is calculated dynamically as projects.data.filter(p => p.status === 'active').length. Contract Value is sum(projects.quotation_amount). Currently, these StatCards are non-interactive summaries.

Proposed Technical Changes:
1. Make 'Active Projects' StatCard clickable, opening a filtered list or modal displaying active projects (code, name, start date, contract value, cost so far).
2. Make 'Contract Value' clickable, triggering a detailed financial modal/drawer showing project-wise breakdown: Quotation Value, Material Cost, Labour Cost, Contractor Cost, Petty Cash Expenses, Loans/Advances, and Net Margin.
Dependencies & Double-Counting Safeguards: Must avoid double-counting transactions between material_receipts (GRN value) and supplier_payments (cash paid out), or attendance (wages earned) and payroll (salary paid).

Complexity: Low–Medium. **Risk:** Low (read-only UI aggregation).

UPDATE 2 — Financial Privacy and Role-Based Dashboard Access

Current State: Database RLS policies for `projects`, `expenses`, `attendance`, and `cash_closing` grant `SELECT` access to all `authenticated` users. Consequently, any logged-in user (`viewer`, `supervisor`, `accounts`, `md`) can view financial totals.

Proposed Technical Changes & Permission Matrix:
The owner requests hiding financial metrics from non-management staff. We propose a strict role matrix where financial totals are suppressed in UI components for `supervisor` and `viewer`, and backend RLS column security or filtered API endpoints prevent direct inspection.

Role Name	Project Info	Financial Figures	Transaction Entry	Approval & Edits	Deletion
Company Owner / MD	Full Access	Full Access	Full Access	Full Approver	Allowed
Admin	Full Access	Full Access	Full Access	Full Approver	Allowed
Accounts Staff	Full Access	Full Access	Enter Payments/Expenses	Edit Drafts	Restricted
Site Supervisor	Assigned Projects	**HIDDEN**	Enter Attendance/Indents	Submit Drafts	Prohibited
Ordinary Employee / Viewer	Read-Only General	**HIDDEN**	Prohibited	Prohibited	Prohibited

UPDATE 3 — Labour Attendance, Wages, and Payroll Removal Evaluation

Current State: Payroll (`src/routes/_authenticated/payroll.tsx`) manages monthly salary runs (`payroll`) and payslip lines (`payroll_items`), calculating gross pay, deductions (PF/ESIC/advances), and net pay with MD approval buttons. Attendance (`src/routes/_authenticated/attendance.tsx`) logs daily attendance hours and daily wages.

Analysis of Alternatives:

- **Option A (Complete Removal of Payroll):** Move advance tracking and weekly settlement directly into Attendance. *Disadvantages:* Destroys monthly statutory compliance records, eliminates payslip generation, loses historical audit trails, and removes formal payroll run approvals.
- **Option B (Integrated Attendance-Payroll Model — RECOMMENDED):** Keep Attendance as the single source of daily wage entry, and auto-populate weekly/monthly settlement reports in Payroll for settlement and disbursement. *Advantages:* Zero data loss, total financial compliance, streamlined UI.

Complexity: High. **Risk:** High if data is deleted; Low if integrated.

UPDATE 4 — Dashboard Spend Breakdown

Current State: StatCard 'Spend (6 months)' sums `expenses.amount` + `attendance.calculated_wage`. It currently represents recorded site operational costs + labour wages, NOT raw material purchases.

Proposed Technical Changes: Make 'Spend' clickable to open a Material Expenditure Breakdown drawer, showing material costs grouped by item/category (e.g. Cement, Steel, Sand, Hardware) sourced from `material_receipts` or tagged `expenses.material_name`. Ensure clear UI distinction between operational expenses and material purchases.

Complexity: Medium. **Risk:** Low (prevents misinterpretation of financial figures).

UPDATE 5 — Office Expenses Section

Current State: `expenses` table supports categories like 'Stationery', 'Telephone/Internet', 'Travel', 'Professional Fees', 'Bank Charges'. Unallocated office costs have `project_id = NULL`.

Proposed Technical Changes: Add a dedicated 'Office Expenses' filter/dashboard widget querying `expenses` where `project_id IS NULL` or category belongs to administrative overhead. This reuses existing schema and avoids creating duplicate transaction structures.

Complexity: Low. **Risk:** Low.

UPDATE 6 — Materials, Suppliers, Contractors & Subcontractors

Current State: Materials and Contractors have separate data workflows. Subcontractors use `work_orders` and `contractor_bills` (lump-sum/unit-rate billing), while daily labour uses `attendance`.

Analysis & Ambiguity Findings:

- *Subcontractor Attendance vs Billing:* Subcontractors are paid per Work Order milestones/measurements, not daily wages. Subcontractor attendance should be tracked as 'Gang Presence' without generating daily wage liabilities.
- *Material Supply Categories:* Notes mention dividing material supply into two types (e.g. Ordinary Suppliers vs Subcontract-inclusive Material Suppliers). Clarification required from management.

Complexity: High. **Risk:** Medium (requires strict separation of contractor billing models).

4. Labour Attendance & Wage Register (Brickwell Specification)

Using the official reference document (**'Brickwell Pvt Ltd 13.09.2026 to 19.09.2026'**), we have audited the exact specifications for the requested weekly construction labour register:

Worker Name	Designation	Daily Wage (■)	Reference Team	7-Day Sample Attendance & Units	Calculated Wage (■)
Vinoth	Mason old	1,100	Vinoth Team	7.5 days (worked overtime)	■8,250
Krishna	Mason	1,100	Vinoth Team	7.0 days	■7,700
Errana	Helper	750	Errana Team	8.0 days (inc. Sunday shift)	■6,000
Ragunath	Helper	750	Errana Team	8.0 days	■6,000
Sanjay	Mason	1,100	Sathish Team	7.0 days	■7,700
Gowtham	Helper	750	Sathish Team	8.0 days	■6,000
Narasamma	Helper	500	Narasamma Team	8.5 days	■4,250
Ravi	Mason	1,100	Vinoth Team	7.0 days	■7,700
Madesh	Helper	750	Errana Team	8.0 days	■6,000
Sathish	Mason	1,100	Sathish Team	7.0 days	■7,700
Beema	Helper	750	Sathish Team	8.0 days	■6,000
Rammapppa	Helper	750	Sathish Team	8.0 days	■6,000

OFFICIAL REGISTER RECONCILIATION TOTALS (13.09.2026 to 19.09.2026):
- ***Total Attendance Units:*** *94.5 Units*
- ***Gross Calculated Wages:*** ■*64,425*
- ***Bus / Petrol Allowances:*** ■*320*
- ***Advances Deducted:*** ■*0*
- ***Net Wage Balance Due:*** ■***64,745*** *(Formula: Calculated Wage + Bus/Petrol - Advance)*

5. Before-and-After Workflow Explanations

To help non-technical management visualize the changes, the table below illustrates how primary operations currently function versus how they will operate post-approval:

Workflow Area	Current Workflow Implementation	Proposed Post-Approval Workflow
Dashboard Financial Drill-down	Static StatCards displaying total active projects, contract value, and spend. No drill-down available.	Authorized users click StatCards to launch detailed financial modals (project breakdown, material expenditure, net margins).
Role-Based Financial Privacy	All logged-in users see financial cards and project values on dashboard and reports.	Strict UI & RLS restrictions hide financial metrics from `supervisor` and `viewer` roles, reserving them for `md`, `admin`, and `accounts`.
Labour Attendance & Weekly Wages	Single daily entry form per worker/contractor. Weekly spreadsheet register not present.	Spreadsheet-like 7-day Weekly Attendance Register with site allocation, fractional units, team totals, allowances, and Saturday payment tracking.
Payroll & Monthly Settlement	Standalone Monthly Payroll module with manual gross/deduction inputs and MD approval.	Integrated Model: Weekly attendance auto-populates monthly settlement runs in Payroll, preserving statutory records and MD approvals.
Office Overhead Expenses	Office costs mixed into site expenses or unallocated records.	Dedicated 'Office Expenses' dashboard widget filtering overheads (`project_id IS NULL`), preventing duplicate transactions.

6. Dependency and Impact Matrix

The matrix below maps each requested update to affected codebase modules, database tables, financial calculations, and security policies:

Req ID	Requirement Description	Affected Code Modules	Affected DB Tables / RLS	Financial / Calculation Impact
UPD-01	Clickable Dashboard Projects & Contract Value	`dashboard.tsx`, `StatCard.tsx`	`projects`, `v_project_summary`	Requires accurate non-double-counted aggregation of GRNs, wages, bills, and expenses.
UPD-02	Financial Privacy & Role-Based Dashboard	`auth.tsx`, `dashboard.tsx`, `reports.tsx`	RLS policies on `projects`, `expenses`, `attendance`	Prevents unauthorized viewing of financial assets, cash on hand, and revenue.
UPD-03	Payroll vs Attendance Restructuring	`attendance.tsx`, `payroll.tsx`	`attendance`, `payroll`, `payroll_items`	Impacts wage calculations, advance recovery, statutory deductions, and payslips.

UPD-04	Clickable Spend & Material Cost Breakdown	`dashboard.tsx`, `materials.tsx`	`material_receipts`, `expenses`	Distinguishes material procurement (GRN) from site operational expenses.
UPD-05	Dedicated Office Expenses Section	`dashboard.tsx`, `accounts.tsx`	`expenses`, `expense_categories`	Isolates corporate administrative costs from direct project expenditure.
UPD-06	Material Suppliers & Subcontractors	`contractors.tsx`, `materials.tsx`	`contractors`, `work_orders`, `suppliers`	Separates piece-rate milestone payments from daily wage attendance.
ATT-01	Weekly Attendance & Wage Register	`attendance.tsx`	`attendance`, `employees`	Standardizes `Wage Balance = Wage + Allowance - Advance` across weekly Saturday runs.

7. Complexity, Risk Summary & Alternatives

Each requested update has been assigned a technical complexity rating based on codebase impact, risk profile, and recommended mitigation:

Req ID	Complexity Level	Identified Technical & Business Risks	Recommended Risk Mitigation Strategy
UPD-01	**Medium**	Double-counting when combining GRN costs and expense lines.	Use explicit SQL views (`v_project_summary`) with strict single-source joins.
UPD-02	**High**	UI hiding without backend RLS enforcement allows API data leaks.	Implement Database RLS policies restricting financial columns by user role.
UPD-03	**Critical**	Deleting Payroll causes loss of statutory records and payslips.	**Adopt Option B:** Integrate Attendance daily entry with Payroll monthly runs.
UPD-04	**Medium**	Confusing general operational spend with material receipts.	Label dashboard cards clearly (e.g. 'Site Spend' vs 'Material GRN Purchases').
UPD-05	**Low**	Creating duplicate expense records for office overheads.	Filter existing `expenses` table where `project_id IS NULL`.
UPD-06	**High**	Misapplying daily wage calculations to milestone subcontracts.	Keep Work Orders (`work_orders`) distinct from daily attendance wages.
ATT-01	**Medium**	Performance degradation when loading 7-day grids for large worker teams.	Use memoized React grid tables with paginated weekly date ranges.

8. Questions Requiring Company Clarification

Before implementation can begin, company management must provide formal written clarification on the following unresolved business rules:

No.	Unresolved Requirement / Ambiguity	Specific Business Decision Needed from Management	Management Decision / Selection
1	**Financial Privacy Access Rights**	Meeting notes mention both 'MD-only' and 'Admin-only'. Exactly which roles are allowed to view financial totals, contract values, and cash on hand?	[ ] MD Only [ ] MD + Admin [ ] MD + Admin + Accounts

2	**Payroll Module Architecture**	Should the Payroll module be completely removed, or integrated with Attendance to preserve monthly payslips, PF/ESIC deductions, and MD approval workflows?	[ ] Remove Payroll [ ] **Integrate with Attendance (Rec.)** [ ] Retain Standalone
3	**Subcontractor Payment Formula**	Are subcontractors paid based on daily attendance hours, work order milestone completion (MB), or a fixed contract amount?	[ ] Work Order Milestones [ ] Daily Attendance [ ] Combination
4	**Material Supply Categories**	Meeting notes refer to dividing material supply into 'two types' and mention the number 'ten'. What does the second type represent?	[ ] Bulk vs Consumable [ ] Supply vs Subcontract [ ] Other: _____
5	**Fractional Attendance & Overtime Rules**	Does 1.5 attendance units represent 1.5 shifts of normal rate, or standard working hours + 1.5x overtime multiplier?	[ ] 1.5 Shifts at Normal Rate [ ] Standard + Overtime Rate [ ] Configurable Rule

9. Company Approval Checklist

Management may individually Approve, Reject, or Request Revision for each item below:

Req ID	Requirement Description	Management Decision	Company Comments / Modification Notes	Approver Signature & Date
UPD-01	Clickable Active Projects & Contract Value Summary	[ ] Approve [ ] Reject [ ] Revise		
UPD-02	Financial Privacy & Role-Based Access Controls	[ ] Approve [ ] Reject [ ] Revise		
UPD-03	Labour Attendance & Payroll Restructuring	[ ] Approve [ ] Reject [ ] Revise		
UPD-04	Clickable Dashboard Spend & Material Breakdown	[ ] Approve [ ] Reject [ ] Revise		
UPD-05	Dedicated Office Overhead Expenses Section	[ ] Approve [ ] Reject [ ] Revise		
UPD-06	Subcontractor & Material Supplier Management	[ ] Approve [ ] Reject [ ] Revise		
ATT-01	Weekly 7-Day Attendance Register & Wage Sheet	[ ] Approve [ ] Reject [ ] Revise		

10. Proposed Implementation Sequence (Post-Approval)

Once management approvals and clarifications are received, implementation will follow a strict 4-phase technical roadmap:

Phase A — Foundation & Security (Week 1): Implement Role-Based Financial Privacy (UPD-02) across RLS policies and UI components. Update role definitions in src/lib/auth.tsx.

Phase B — Attendance & Payroll Integration (Week 2): Build the spreadsheet-like 7-Day Weekly Attendance Register (ATT-01). Connect wage calculations directly to monthly Payroll runs (UPD-03).

Phase C — Subcontractor & Procurement Workflows (Week 3): Enhance Subcontractor Work Order tracking (UPD-06) and configure Office Overhead filtering (UPD-05).

Phase D — Interactive Dashboard & Drill-downs (Week 4): Connect clickable StatCards (UPD-01, UPD-04) for Active Projects, Contract Values, and Material Spend breakdowns.

11. Technical Evidence & Verification Log

This audit was performed strictly on the local repository (Corpus: Ajaykannagit/track-site). Key inspected files and artifacts:

• src/lib/auth.tsx — Authentication context & role permission flags (`isMD`, `canWriteFinance`, `canWriteSite`).

• src/routes/_authenticated/dashboard.tsx — Dashboard metrics and StatCard components.

• src/routes/_authenticated/attendance.tsx — Daily attendance logging & wage calculations.

• src/routes/_authenticated/payroll.tsx — Payroll runs, payslip lines & MD approval logic.

• supabase/migrations/20260918040000_production_enhancements.sql — Database helper views (`v_project_summary`, `v_contractor_outstanding`).

REPORT END — BRICKWELL PVT. LTD. MANAGEMENT AUDIT